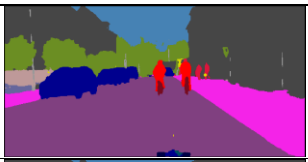
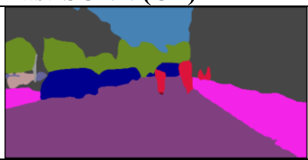
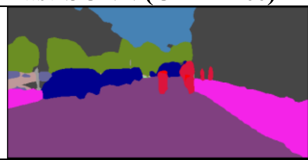
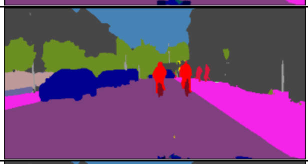
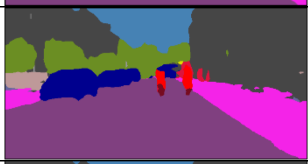
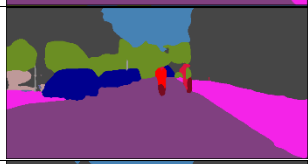
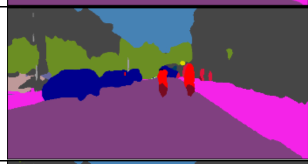
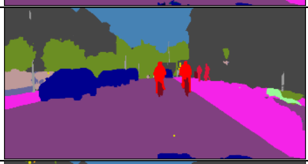
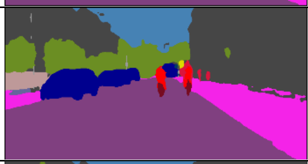
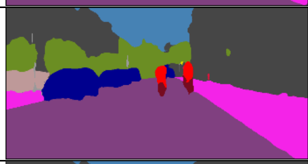
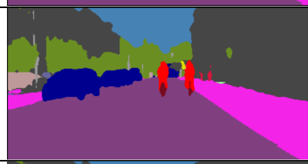
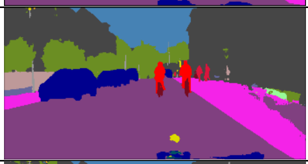
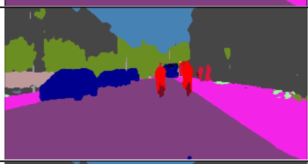
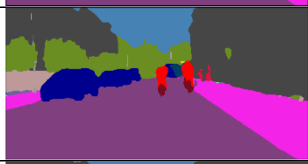
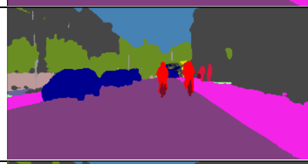
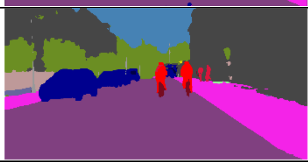
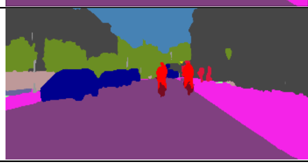
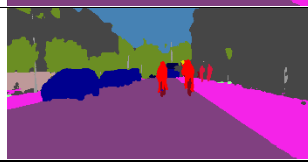
Predictions from each neural network architecture for specific urban scene at different input resolutions

Figure 1 shows a specific urban scene with the semantic segmentation ground-truth labels.

Figure 1. Semantic segmentation ground-truth labels

Table 1 shows the predictions from each neural network architecture for this specific urban scene at different input resolutions.

Table 1. Predictions from each neural network architecture at different input resolutions

S	U-Net baseline model	Fast-SCNN (CL)	Fast-SCNN (CL + KD)	Fast-SCNN (CL + Dice)
1				
2				
3				
4				
5				

The U-Net model architecture recognizes the bicycles and the leaves of the tree on the right in the first curriculum stage (lowest input resolution).

The Fast-SCNN model architecture does not detect the bicycles and the leaves of the tree on the right in the first curriculum stage (lowest input resolution). These urban elements are recognized in the later curriculum stages (higher input resolutions).

The Fast-SCNN model architecture that was trained with Curriculum Learning and Knowledge Distillation has a reduced performance and efficiency. The 2 people in the back on the sidewalk are not detected in the first curriculum stages.
We had no time to further fine-tune the alpha and temperature parameters, in order to get a better performance and efficiency.
The alpha parameter controls the balance between learning from the teacher's outputs and the true labels.
The temperature parameter softens the teacher's predictions to reveal inter-class relationships.

The Fast-SCNN model provides a much better efficiency than the U-Net baseline model for all respective curriculum stages, at the cost of a reduced but acceptable performance.

Fast-SCNN balances performance and computational cost, making it suitable for scenarios where both responsiveness and accuracy are critical. With fewer parameters and reduced computation, it is well-suited for embedded systems.

The Fast-SCNN model is superior: we reach the highest efficiency in the TUE CodaLab rankings of 2025 and 2024.